

Autonomous Causal Discovery: Evaluating LLMs' Priors and Constraint Strategies for Reliability

Lyuzhou Chen¹, Xiangyu Wang¹, *Member, IEEE*, Taiyu Ban¹, Derui Lyu¹, Qinrui Zhu¹, Xin Wang,
and Huanhuan Chen¹, *Fellow, IEEE*

Abstract—Expert-guided Causal Structure Learning (CSL) incorporates prior knowledge to improve the accuracy of causal discovery, yet the acquisition of such knowledge is often restricted by the availability of human experts. While Large Language Models (LLMs) provide an alternative source of causal priors, LLM-derived knowledge can be inconsistent with the true causal structure due to hallucinations or contextual misinterpretations. This paper introduces a structural constraint measurement framework, which defines constraint strength and constraint quality to describe reliability and effectiveness, enabling a systematic evaluation of LLM-derived constraints. Using this framework, we evaluate five categories of structural constraints: Edge Existence (EEC), Edge Forbidden (EFC), Path Existence (PEC), Path Forbidden (PFC), and Order Constraints (OC). Our theoretical and empirical analyses demonstrate that while EEC offers high constraint strength, it exhibits low quality when derived from LLMs; conversely, PFC and OC provide a balanced trade-off between search-space pruning and reliability. Building on these insights, we propose a two-level CSL optimization framework that partitions the search space by node order and refines the structure using global path constraints. The results show that this framework provides an effective way to incorporate noisy LLM-derived priors into CSL, particularly in settings where expert knowledge is limited.

Index Terms—Causal structure learning, large language models, structure constraints.

I. INTRODUCTION

RECOVERING the actual causal mechanisms from real-world observations is challenging due to data limitations [1], [2], where expert guidance is essential for discovering insightful and meaningful causal relationships [3], [4]. Recently, the expert-guided Causal Structure Learning (CSL) [5], [6], [7], [8], [9], [10], using prior-based structure learning technique, has raised emerging research attention in a wide range

of domains [11], [12]. This approach effectively enhances the reliability of discoveries by relying on the precise analysis by experts on the exact structure-level causality [13], [14]. However, it struggles to be widely applicable due to the demand for costly expert resources [15], [16], [17]. Although some efforts have attempted to explore more general, automated methods for acquiring priors, such as through knowledge bases and webpage texts, the reliability of these priors is often difficult to ensure [18], [19]. Therefore, exploring more accessible methods for acquiring priors and more reliable modes for applying them is key to overcoming practical bottlenecks in CSL.

The success of Large Language Models (LLMs) in intelligent reasoning [20], [21], [22], [23] has opened a new frontier for the autonomous prior acquisition, providing knowledge spanning a wide range of domains [24], [25], [26]. This automated, end-to-end approach can provide vast amounts of human-cognition-consistent, context-specific causal knowledge at low cost, which can then be converted into constraints of different forms. This approach does not require expert involvement and can expand the practical application scope of CSL. Therefore, LLM-based prior knowledge acquisition quickly gains attention in the CSL community and promises to be the next mainstream method [27].

Despite their promising application potential, current LLMs may encounter reliability issues due to hallucinations or context misinterpretation [28], [29]. As illustrated in Fig. 1, if CSL algorithms incorporate incorrect LLM-derived knowledge, the resulting graph may deviate substantially from the true causal structure. This issue highlights a fundamental difference between traditional expert-guided CSL and LLM-based autonomous prior acquisition.

Fig. 1 further illustrates that prior knowledge often acts through structural constraints on causal relationships, such as the existence of edges or paths. In addition, the same piece of knowledge can be translated into different forms of constraints. For example, edge constraints can also be interpreted as path constraints. This observation leads to a pivotal trade-off between reliability and efficiency. Using a looser constraint strategy better preserves reliability when background exploration is insufficient, yet it may result in the loss of pruning efficacy.¹ [30]. Specifically, recent studies indicate that LLMs often struggle to distinguish between direct and indirect causal relationships [31], and their ability to generalize to rare or novel relations remains

Received 2 December 2024; revised 24 March 2026; accepted 24 April 2026. Date of publication 4 May 2026; date of current version 6 August 2026. This work was supported in part by the National Nature Science Foundation of China under Grant 62406302, Grant 62137002, and Grant 62576327, in part by the Natural Science Foundation of Anhui province under Grant 2408085QF195, in part by the USTC Research Funds of the Double First-Class Initiative under Grant YD9110002085, and in part by the Research Funds of Centre for Leading Medicine and Advanced Technologies of IHM under Grant 2025IHM01030. Recommended for acceptance by L. Shen. (Corresponding authors: Xiangyu Wang; Huanhuan Chen.)

The authors are with the School of Computer Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: sa312@ustc.edu.cn; hchen@ustc.edu.cn).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TPAMI.2026.3689960>, provided by the authors.

Digital Object Identifier 10.1109/TPAMI.2026.3689960

¹For example, the edge constraint $x \rightarrow y$ enforces the direct causal relation that ‘ x directly causes y ’, whereas the path constraint $x \rightsquigarrow y$ only requires an ancestral relation and does not enforce direct causation.

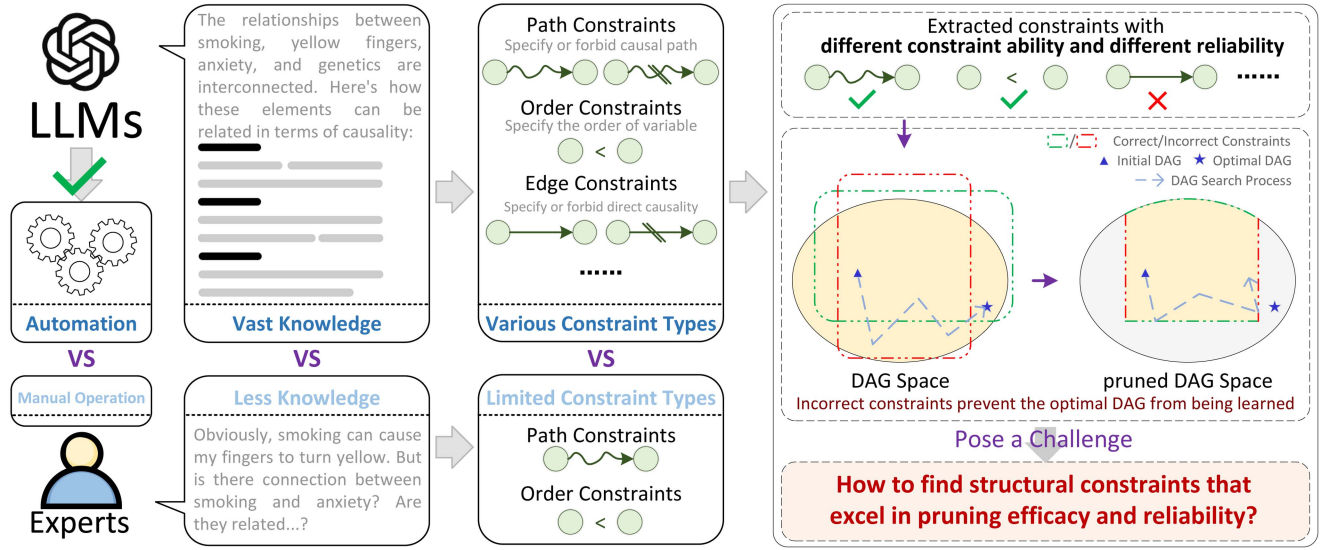


Fig. 1. Comparison of the experts-based and LLMs-based approaches to acquiring causal knowledge and the potential challenges of using LLMs.

limited [32]. Related limitations of LLM-based causal reasoning have also been discussed in [33]. In such scenarios, overly strict constraints can prematurely eliminate the ground-truth structure once the identified prior is incorrect, whereas looser constraints preserve more candidate structures and are therefore more robust to prior uncertainty. Consequently, the central research goal of this paper is to identify which types of constraints best balance between reliability and efficiency, as posed in Fig. 1. Current research has mainly focused on the application of specific priors and rarely provides a systematic discussion of how different constraint types mediate this trade-off.

To address this issue, this paper focuses on the two stages of constraint acquisition and application, establishes a theoretical structural constraint measurement framework to evaluate the overall performance of structural constraints on CSL, and then proposes a framework for the use of reliable prior knowledge.

Specifically, this paper first defines two crucial metrics, constraint quality, and constraint strength, based on the theoretical lower bound of the accuracy of LLM-derived constraints and their effectiveness in screening the potential causal structure, respectively. Using these metrics, this paper then theoretically evaluates the five widely used structural constraints to obtain their compatibility with LLMs. Based on the evaluation results, this paper points out the structural constraints that are suitable for automated acquisition and utilization. Some of these constraints may be easily overlooked in traditional expert-guided methods because the high reliability of expert knowledge requires only efficient constraints to be considered. However, in the context of LLM-based prior acquisition, structural constraints with low efficiency but high reliability also play a role. Furthermore, based on the characteristics of different constraints, this paper proposes a reliability-driven CSL framework for the systematic integration of LLM-derived knowledge. By utilizing constraints that excel in both acquisition and effectiveness, this framework enhances the automation and reliability of CSL through a coordinated integration mechanism, thereby promoting the application

of causal methods even when high-reliability expert knowledge is insufficient. The contributions of this paper are as follows:

- The establishment of a structural constraint measurement framework, including the metrics of constraint strength and constraint quality, evaluating the constraints used in LLM-based CSL.
- The assessment of the compatibility between widely-used structural constraints and LLMs, highlighting the potential value of some constraints that have been overlooked in traditional research.
- The proposal of a reliability-oriented CSL integration framework for the LLM setting, which systematically incorporates structural constraints with favorable acquisition quality and practical utility.

II. RELATED WORK

This section introduces recent research on using LLMs to discover causal relationships from two perspectives. First, Section II-A discusses the application of LLMs as a new method for causal knowledge acquisition. Then, Section II-B examines the capabilities of LLMs in causal relationships, including their strengths and limitations.

A. Applications of LLMs in Causal Knowledge Exploration

Recent research has evolved from traditional expert-guided CSL to autonomous frameworks that utilize LLMs for causal knowledge acquisition and structural optimization [16]. Unlike traditional methods that rely on human-defined edge or order constraints [3], [34], LLM-based approaches can be categorized into three main strategies: prior-based acquisition, LLM-guided search, and representation-based approaches.

a) Prior-based Methods: This category focuses on extracting structured causal information from LLMs to serve as static inputs for CSL algorithms. For instance, LLMs have been utilized to answer conditional independence queries, which are then

processed by the PC algorithm [35] to reconstruct the DAG [36]. Other studies have prompted LLMs to identify root nodes or ancestral relationships [37]. These methods focus on turning the model's broad internal knowledge into specific structural priors. However, most prior-based methods treat extracted priors as directly usable inputs, with limited analysis of how prior-type reliability affects downstream CSL.

b) LLM-guided Search and Refinement: More recent frameworks have integrated LLMs directly into the optimization to guide the search process dynamically. Rather than providing a one-time prior, LLMs can be used to validate edges generated during each iteration of a data-driven search, effectively pruning the search space in real-time [38]. Furthermore, some approaches employ LLMs as post-hoc refiners to evaluate and correct the output of score-based algorithms, ensuring that the final structure aligns with both data and domain common sense [39]. Such strategies often incorporate credibility scores or self-reported confidence from LLMs to manage the weight of the provided guidance [40]. These methods emphasize dynamic LLM-search interaction, but typically rely on empirical heuristics rather than theoretically characterizing the structural risk of different constraint types.

c) Representation and Semantic Approaches: Beyond structural constraints, LLMs contribute to causal discovery through their ability to understand and represent complex variable semantics. By analyzing variable descriptions and metadata, LLMs can identify potential causal pathways that might be overlooked by purely statistical methods, especially in high-dimensional or medical domains where variable names carry significant contextual weight [38], [41]. Unlike prior-based or search-guided methods, this line of work focuses on semantic representations rather than on the reliability of structural constraint forms for downstream integration. For more detailed information, please refer to relevant reviews [42], [43], [44].

Beyond CSL-oriented methods, recent work has also explored the combination of LLMs and causal graphs for scientific hypothesis generation. For example, Tong et al. [45] used LLMs to extract causal relation pairs from a large-scale psychology corpus, constructed a domain-specific causal graph, and generated candidate hypotheses through link prediction. This line of research is closely related in spirit to our work, while addressing a different application setting from CSL.

B. Hallucination Mitigation in LLM-Based Causal Discovery

The causal perception and reasoning capabilities of LLMs are highly controversial. While some studies demonstrate that LLMs can achieve state-of-the-art accuracy on specific causal benchmarks and accurately predict edge directions in domain-specific medical data [41], [46], these successes are often constrained by the models' inherent limitations in high-precision logical derivation [47] and their tendency to recite training data rather than perform genuine causal exploration in new scenarios [33], [48]. Other research also confirmed similar observations [49], [50].

A critical bottleneck for autonomous CSL is the phenomenon of hallucinations, where LLMs generate causal relationships

TABLE I
COMPARISON OF TYPICAL LLM-BASED CAUSAL DISCOVERY METHODS

Research	Structural Constraints	Hallucination Mitigation	Decision Authority
[16]	Edge	Empirical Quantification	Model-Decisive
[25]	Edge	Self-Consistency	Data-Centric
[27]	Edge	Majority Voting	Knowledge-Centric
[30]	Ancestral	Empirical Quantification	Integrative Synergy
[31]	Order	Majority Voting	Integrative Synergy
[33]	Edge	Adversarial Semantic Probing	Model-Decisive
[36]	Edge	Majority Voting	Model-Decisive
[37]	Edge	Multi-turn Context	Model-Decisive
[39]	Edge	Self-reported Confidence	Model-Decisive
[38]	Edge	Reflexive Memory	Integrative Synergy
[40]	Edge	Self-reported Confidence	Integrative Synergy
[46]	Edge	Empirical Quantification	Model-Decisive
[41]	Edge	Human Knowledge	Model-Decisive
[48]	Edge	Contextual Grounding	Model-Decisive
[50]	Edge	Human Knowledge	Model-Decisive
Ours	Multi-level	Theoretical proof & Empirical Quantification	Integrative Synergy

that extend beyond their actual knowledge boundaries [51]. If LLM-derived knowledge is incorporated into the CSL algorithm without reliability control, such hallucinations can lead to significant structural distortions, causing the resulting DAG to deviate substantially from the ground truth. This gap highlights a difference between traditional expert-guided CSL and LLM-based autonomous acquisition. To mitigate these risks, researchers have developed various strategies for LLM-derived priors, as detailed in Table I. However, these strategies mainly provide empirical improvements and often do not theoretically quantify the structural risk associated with different types of priors.

Facing this challenge, this paper establishes a structural constraint measurement framework to characterize the reliability of LLM-derived knowledge. Our theoretical and empirical analyses reveal that path forbidden and order constraints provide a superior trade-off between search-space pruning and reliability. By identifying and utilizing high-reliability priors, our approach allows for a systematic integration of knowledge that effectively mitigates the uncertainty inherent in autonomous discovery processes.

III. PRELIMINARY

Definition 1 (Bayesian Network): A Bayesian network (BN) [52] (G, p) models the joint probability distributions among random variables $X = \{x_1, x_2, \dots, x_n\}$, where directed acyclic graph (DAG) $G = (X, E)$ represents the conditional dependence between variables in X . Joint probability distribution p satisfies the Markov property and can be decomposed into the product of local conditional independent distributions [53]:

$$p(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | pa(x_i)) \quad (1)$$

where $pa(x_i)$ represents the set of parent nodes of x_i in graph G , which is indicated according to edges set $E \subset X \times X$.

The structure of DAG G naturally describes causal relationships among the variables of interest, with each edge representing a direct causal effect from the parent variable to the child variable. Paths with multiple edges indicate that the source variable is a distant cause of the target variable. This paper also refers to random variables as nodes in the graph G when there is no confusion.

Definition 2 (Causal Structure Learning): Let D denote the dataset, where each sample is independently drawn from the joint distribution p . The objective of causal structure learning is to find a suitable DAG such that there exists a corresponding local conditional distribution $P(x_i|pa(x_i))$ whose product is consistent with the joint distribution expressed by D .

Extensive research has explored the use of prior knowledge to construct structural constraints for guiding the structural learning of DAGs. Under such constraints, CSL must not only identify a DAG that fits the data D but also ensure it adheres to the specified priors. Based on existing studies [16], [54], mainstream causal structural constraints can be categorized into the following five types:

Definition 3 (Structure Constraints): The structure constraints on the DAG mainly include:

- a) *Edge Existence Constraint (EEC)*: Specifies that node x is the parent of the node y , denoted as $x \rightarrow y$.
- b) *Edge Forbidden Constraint (EFC)*: Specifies that node x is not the parent of the node y , denoted as $x \nrightarrow y$.
- c) *Path Existence Constraint (PEC)*: There exists a path from node x to node y within the DAG, denoted as $x \rightsquigarrow y$.
- e) *Path Forbidden Constraint (PFC)*: There is no path from node x to node y within the DAG, denoted as $x \nrightsquigarrow y$.
- d) *Order Constraint (OC)*: Specifies that node x precedes node y in the topological ordering, denoted as $x < y$.

In terms of the scope of influence, the impact of EECs and EFCs is local, whereas the impact of PECs and OCs is global. Local constraints are relatively easier to handle because they only need to be considered when changes occur in the relevant structures. In contrast, satisfying global constraints depends on the overall structure of the DAG. Therefore, when considering algorithms that use global constraints, constructing a framework that can effectively handle these constraints is key to improving efficiency.

Regarding the frequency of use, most research focuses on local constraints, with considerably less attention given to the properties and applications of global constraints. Among these, PFCs have been particularly underexplored. Intuitively, PFCs and PECs share similarities in their definitions and properties and represent a natural type of constraint. However, these two constraints differ significantly in how they are derived and processed. These differences will be discussed in detail in Section IV.

To facilitate subsequent discussion, this section finally defines constraint space:

Definition 4 (DAG Space and Constraint Space): Let $\mathcal{G}$ denote the space of DAGs with n nodes, where each element represents a possible DAG. Subscripts are utilized to distinguish subspaces of $\mathcal{G}$, i.e. constraint space, under different constraints.

For example, $\mathcal{G}_{A \rightarrow B}$ denotes the set of all DAGs that satisfy the constraint $A \rightarrow B$.

Constraint space represents the set of all DAGs that satisfy the constraints, which is important for discussing constraint strength.

IV. STRUCTURAL CONSTRAINT THEORETICAL MEASUREMENT FRAMEWORK

Different constraints differ in reliability and effectiveness in screening the potential causal structure. To reasonably evaluate the performance of structural constraints given by LLM, this paper constructs a structural constraint measurement framework. This section first describes two evaluation perspectives in Section IV-A, including constraint strength and constraint quality. Subsequently, Sections IV-B and IV-C delve into the constraint strength of widely used structures constraints, while Section IV-D involves a theoretical analysis of the constraint quality of these constraints. Based on these analyses, constraints that are suitable to be used will be identified. Furthermore, a CSL framework that can integrate and fully utilize suitable constraints will be presented in Section V.

A. Constraint Strength and Constraint Quality

This section defines constraint strength and constraint quality, the former quantitatively determining how much information a constraint can provide, and the latter characterizing the degree of reliability of the LLMs-derived constraints.

Definition 5: The strength constraint CS of a constraint c is the amount of reduction in the uncertainty of the DAG.

$$CS(c) = H(\mathcal{G}) - H(\mathcal{G}_c) \\ = - \sum_{g \in \mathcal{G}} P(g) \log P(g) + \sum_{g \in \mathcal{G}_c} P(g) \log P(g) \quad (2)$$

where $H(\mathcal{G})$ and $H(\mathcal{G}_c)$ reflect the uncertainty when randomly selecting a DAG from the space $\mathcal{G}$ and $\mathcal{G}_c$, respectively. Assuming that each possible DAG has the same prior probability, the constraint strength can be simplified to:

$$CS(c) = \log |\mathcal{G}| - \log |\mathcal{G}_c| \quad (3)$$

Given the definition of constraint strength, its calculation is quite complex, as it requires counting the number of DAGs that satisfy the constraint. Fortunately, for common structural constraints which are described in Section III, there is a more convenient method to assess constraint strength, which will be discussed in detail in Sections IV-B and IV-C.

In many traditional expert-guided researches, the quality issues of constraints are generally ignored. However, for LLMs whose behaviors are not guaranteed, the quality of constraints requires close attention. To achieve this, this paper first formally defines the behavior of LLMs. Specifically, the following assumption is proposed first:

Assumption 1: When determining the causal relationship between variables, an LLM can be regarded as a classifier $M_I : X \times X \rightarrow Y$, where I represents the input instruction, X

TABLE II
SUMMARY OF THE BENCHMARK NETWORKS USED IN THE EXPERIMENTS

Network	#Nodes	#Arcs	Maximum in-degree	#Parameters	Application domain
ASIA	8	8	2	18	A benchmark network for tuberculosis with <i>Asia</i> travel as a risk factor
LUCAS	12	12	2	56	A benchmark network for <i>Lung Cancer dataSet</i>
CHILD	20	25	2	230	A benchmark network for <i>child</i> diseases
NEURO	26	26	2	106	A benchmark network for <i>neuropathic</i> pain diagnosis
INSURANCE	27	52	3	1008	A benchmark network modeling claim costs for <i>insurance</i> policyholder
WATER	32	66	5	10083	A benchmark network for <i>wastewater</i> treatment control systems
MILDEW	35	46	3	540150	A benchmark network for winter wheat <i>mildew</i> control
ALARM	37	46	4	509	A benchmark network developed for <i>alarm</i> message systems

* '#Nodes', '#Arcs', and '#Parameters' represent the number of nodes, edges, and parameters of the ground truth graph structure corresponding to the dataset, respectively.

is the set of random variables, and Y is the space of possible outcomes.

For example, I could be “Please identify the causal relationships among the following variables”. Note that, unlike traditional classifiers, the output space of LLMs is extensive and complex, often resulting in long text outputs. Therefore, classifier M_I is supposed to contain an implicit post-processing step to transform the LLMs’ output into an appropriate format. Different instructions I will trigger different model behaviors, and therefore, different I correspond to different M_I . This paper sets the output space $Y = \{0, 1\}$, where $M_I(x_i, x_j) = 1$ indicates that the LLM affirmatively asserts the queried structural property, and $M_I(x_i, x_j) = 0$ indicates that it does not. The corresponding ground-truth label $\hat{y} = 1$ means that the queried property is indeed true for the variable pair. Under this convention, the queried property may be either positive in semantics (e.g., the existence of an edge in EEC) or negative in semantics (e.g., the absence of a path in PFC). Therefore, Definition 6 consistently measures the precision of the asserted property across different types of structural constraints.

Based on the above assumptions, the definition of constraint quality is:

Definition 6: The constraint quality CQ of a type of constraint c is the precision of its acquisition method M_I :

$$CQ(c) = P(M_I(x_i, x_j) = \hat{y} | M_I(x_i, x_j) = 1) \quad (4)$$

where $\hat{y}$ represents the ground truth of the relationship between x_i and x_j .

The above formula formally expresses the precision of variable pairs with specified relationships given by LLMs under instruction I . Similar to the precision metric in standard classification tasks, the definition of CQ focuses exclusively on the alignment between the acquired constraints and the ground truth $\hat{y}$. While the acquisition method M_I involves LLMs, we treat the specific model choice, the prompt design, and the post-processing and parsing procedures as external procedural factors that influence the final CQ , rather than as part of the intrinsic definition itself. For the constraint quality of specific structural constraints, this paper will conduct a theoretical analysis in Section IV-D.

B. Constraint Strength Assessment of Structural Constraints

This section explores the strength of common structural constraints based on their defined strengths, focusing particularly on ranking these constraints in terms of their relative strength.

Constraint strength is defined in terms of its magnitude relative to the capacity of the constraint space, which can be complex to calculate. However, in practice, it’s not always necessary to determine the exact capacity of each constraint space. For example, when one constraint space includes another, it indicates that the constraint with the larger space has weaker strength because it imposes fewer restrictions. This inclusion relationship suggests that one constraint is a prerequisite for the other, simplifying the comparison of strengths. Building on this understanding, we will demonstrate through a chain-like containment relationship among the five types of structural constraints defined in Section III, which establishes a hierarchy of constraint strength. This approach not only elucidates the theoretical underpinnings but also aids in practical applications by simplifying the evaluation of constraint spaces.

Some readily observable relationships between constraint spaces are discussed first. Since satisfying the edge constraint $A \rightarrow B$ also implies the existence of the path $A \rightsquigarrow B$, it follows that any DAG satisfying the edge constraint $A \rightarrow B$ also satisfies the path constraint $A \rightsquigarrow B$. This indicates that $\mathcal{G}_{A \rightarrow B} \subseteq \mathcal{G}_{A \rightsquigarrow B}$. Similarly, DAGs that violate the edge constraint $A \not\rightarrow B$ also violate the PFC $A \not\rightsquigarrow B$, thus $\mathcal{G}_{A \not\rightarrow B} \subseteq \mathcal{G}_{A \not\rightsquigarrow B}$.

Moreover, according to the acyclicity of DAG, satisfying $A \rightsquigarrow B$ necessarily means that $B \not\rightsquigarrow A$ holds. Conversely, $B \not\rightsquigarrow A$ does not necessarily imply $A \rightsquigarrow B$ (consider two independent nodes). Therefore, we establish that $\mathcal{G}_{A \rightsquigarrow B} \subseteq \mathcal{G}_{B \not\rightsquigarrow A}$. In summary, the following hierarchy can be concluded:

$$\mathcal{G}_{A \rightarrow B} \subseteq \mathcal{G}_{A \rightsquigarrow B} \subseteq \mathcal{G}_{B \not\rightsquigarrow A} \subseteq \mathcal{G}_{B \not\rightarrow A} \quad (5)$$

which illustrates the strength ranking of EEC, EFC, PEC, and PFC.

Finally, we consider the strength of OCs. Clearly, a DAG that satisfies the path constraint $A \rightsquigarrow B$ necessarily satisfies the OC $A < B$. However, it is not easy to see which of the OC $A < B$ and the PFC $B \not\rightsquigarrow A$ is stronger. In fact, OC $A < B$ contains multiple topological ordering, and each topological

ordering contains multiple DAGs. Whether these DAGs satisfy PFC $B \not\rightsquigarrow A$ requires careful discussion.

To discuss this problem, we first clearly define the constraint space of OCs. A mapping $\Psi : \mathcal{O} \rightarrow 2^{\mathcal{G}}$ from the node order space to the power set of the DAG space is used to represent the DAG that the node order may correspond to, where $\mathcal{O}$ represents the node order space where each element is a topological ordering of nodes. Let $\mathcal{O}_{A < B}$ represent the set of topological orderings in $\mathcal{O}$ that satisfy the OC $A < B$. Then, the constraint space consistent with OC $A < B$ can be represented as:

$$\mathcal{G}_{A < B} = \Psi(\mathcal{O}_{A < B}) \quad (6)$$

Based on these definitions, this paper proposes the following lemma:

Lemma 1 (A single path forbidden constraint is equivalent to a order constraint): The DAG space consistent with a path forbidden constraint $A \not\rightsquigarrow B$ is equivalent to that consistent with the OC $B < A$, i.e.:

$$\Psi(\mathcal{O}_{B < A}) = \mathcal{G}_{A \not\rightsquigarrow B} \quad (7)$$

The proof of Lemma 1 is detailed in the following subsection. This lemma demonstrates that, in certain cases, OCs and PFCs have equivalent constraint strength and can be used interchangeably. However, there are notable differences between these two types of constraints. To explore this further, we present the next lemma:

Lemma 2: The DAG space of the set of OCs is a subset of that of the set of path forbidden constraints, i.e.:

$$\Psi(\cap_{o_i \in S(<)} \mathcal{O}_{o_i}) \subseteq \cap_{p_i \in S(\not\rightsquigarrow)} \mathcal{G}_{p_i} \quad (8)$$

where $S(<)$ represents the set of OCs, and $S(\not\rightsquigarrow)$ represents the set of PFCs, and the constraints in $S(<)$ and $S(\not\rightsquigarrow)$ are in one-to-one correspondence.

According to Lemma 1 and Lemma 2, it can be obtained that:

$$\begin{aligned} \mathcal{G}_{A \rightarrow B} &\subseteq \mathcal{G}_{A \rightsquigarrow B} \subseteq \mathcal{G}_{A < B} = \mathcal{G}_{B \not\rightsquigarrow A} \subseteq \mathcal{G}_{B \not\rightarrow A} \\ \mathcal{G}_{S(\rightarrow)} &\subseteq \mathcal{G}_{S(\rightsquigarrow)} \subseteq \mathcal{G}_{S(<)} \subseteq \mathcal{G}_{S(\not\rightsquigarrow)} \subseteq \mathcal{G}_{S(\not\rightarrow)} \end{aligned} \quad (9)$$

Where $S(\rightarrow)$, $S(\rightsquigarrow)$, $S(<)$, $S(\not\rightsquigarrow)$, $S(\not\rightarrow)$ are sets consisting of corresponding constraints of EEC, PEC, OC, EFC, PFC. In summary, this paper has successfully ranked the five structural constraints in terms of strength:

$$\begin{aligned} CS(S(\rightarrow)) &> CS(S(\rightsquigarrow)) > CS(S(<)) \\ &> CS(S(\not\rightsquigarrow)) > CS(S(\not\rightarrow)) \end{aligned} \quad (10)$$

It can be seen that EEC is the strongest in terms of constraint strength, while the three types of global constraints fall in the middle. However, this does not necessarily mean that EEC performs best in real-world scenarios. The precision of the constraints is also a critical factor to consider. If the constraints are inaccurate, the constrained space may exclude the true DAG, leading to candidate DAGs that deviate significantly from the actual structure. Thus, the stronger the constraint, the more essential it becomes to ensure the correctness of the corresponding prior knowledge.

TABLE III
PERFORMANCE OF DIFFERENT STRUCTURAL CONSTRAINTS GIVEN BY LLM

Network	Metric	Constraint Type				
		EEC	PEC	OC	PFC	EFC
ASIA	Precision	0.50	0.80	1.00	0.85	0.93
	Recall	0.88	0.67	0.71	0.76	0.70
	F1	0.64	0.73	0.83	0.81	0.80
	#Constraints	14	15	27	34	42
LUCAS	Precision	0.43	0.55	0.77	0.86	0.98
	Recall	1.00	0.56	0.90	0.66	0.66
	F1	0.60	0.55	0.83	0.75	0.79
	#Constraints	28	33	117	77	89
CHILD	Precision	0.20	0.25	0.85	0.89	0.96
	Recall	0.76	0.61	0.80	0.58	0.62
	F1	0.32	0.36	0.82	0.70	0.75
	#Constraints	95	155	300	207	240
NEURO	Precision	0.11	0.15	0.89	0.99	1.00
	Recall	1.00	0.94	0.66	0.55	0.56
	F1	0.19	0.26	0.76	0.71	0.71
	#Constraints	241	308	444	334	361
INSURANCE	Precision	0.29	0.50	0.88	0.82	0.97
	Recall	0.79	0.65	0.84	0.84	0.81
	F1	0.42	0.57	0.86	0.83	0.88
	#Constraints	143	217	510	543	560
WATER	Precision	0.26	0.42	1.00	0.86	0.95
	Recall	0.39	0.32	0.59	0.67	0.72
	F1	0.32	0.36	0.74	0.75	0.82
	#Constraints	99	132	483	642	729
MILDEW	Precision	0.24	0.53	0.91	0.83	1.00
	Recall	0.78	0.47	0.73	0.80	0.75
	F1	0.36	0.50	0.81	0.82	0.85
	#Constraints	153	253	727	872	883
ALARM	Precision	0.21	0.37	0.84	0.86	0.99
	Recall	0.67	0.58	0.82	0.70	0.74
	F1	0.32	0.45	0.83	0.77	0.84
	#Constraints	150	347	1077	897	987
Average	Precision	0.36	0.47	0.90	0.87	0.97
	Recall	0.81	0.63	0.77	0.72	0.70
	F1	0.46	0.51	0.83	0.78	0.81

* '#Constraints' represents the number of acquired constraints. 'Average' represents the average metric of 8 causal networks. The metrics exceeding 0.90 are bolded, indicating that the corresponding structural constraints given by LLM on this network are reliable.

C. Detailed Elaboration of the Conclusion in Constraint Strength Assessment

Before presenting the formal proof, we first summarize the key intuition behind Lemma 1 and Lemma 2. The core idea is to establish a bidirectional correspondence between path and order. We show that any DAG consistent with a single node order constraint necessarily satisfies the corresponding PFC, and conversely, any DAG with a forbidden path can be mapped to at least one compatible topological ordering.

To formalize this, we first introduce the mapping $\Phi : \mathcal{G} \rightarrow 2^{\mathcal{O}}$, which is the inverse function of mapping Ψ , representing the possible node topological orderings of a DAG. Furthermore, let $\Phi(\mathcal{H})$ represent the union of node topological orderings corresponding to all DAGs in $\mathcal{H}$ for the sake of description,

TABLE IV
CSL PERFORMANCE BASED ON DIFFERENT PROPORTIONS OF PATH EXISTENCE CONSTRAINTS AND PATH FORBIDDEN CONSTRAINTS

Network	DataSize	Structure Constraint Proportion (PEC, PFC)											
		0, 0		20%, 0		0, 20%		40%, 0		20%, 20%		0, 40%	
		SHD	F1	SHD	F1	SHD	F1	SHD	F1	SHD	F1	SHD	F1
ASIA	250	3.1	0.75	3.0	0.76	1.8	0.87	3.1	0.74	1.8	0.87	1.8	0.88
	1000	1.6	0.86	1.8	0.84	1.1	0.92	1.8	0.83	1.2	0.92	1.1	0.93
LUCAS	250	4.0	0.75	2.9	0.85	3.6	0.77	2.7	0.85	2.0	0.90	2.1	0.89
	1000	2.0	0.85	1.2	0.91	1.2	0.90	1.1	0.92	0.6	0.96	0.1	1.00
CHILD	500	8.3	0.76	7.7	0.78	6.6	0.82	7.2	0.80	6.6	0.82	6.4	0.83
	2000	3.4	0.88	2.8	0.91	2.1	0.94	2.3	0.93	1.7	0.95	1.6	0.96
NEURO	500	51.4	0.13	50.9	0.14	48.3	0.14	50.7	0.14	43.1	0.19	44.4	0.11
	2000	52.2	0.16	52.0	0.16	50.2	0.15	51.9	0.16	43.6	0.20	45.1	0.15
INSURANCE	500	33.2	0.51	30.3	0.59	27.3	0.64	29.5	0.61	27.2	0.64	25.9	0.67
	2000	29.1	0.58	19.4	0.76	17.3	0.79	19.0	0.77	16.8	0.80	15.8	0.82
WATER	1000	54.0	0.32	48.7	0.41	49.8	0.39	46.1	0.46	46.9	0.45	47.4	0.43
	4000	50.0	0.40	47.4	0.44	44.4	0.48	36.8	0.61	43.8	0.50	38.0	0.59
MILDEW	1000	43.3	0.21	39.1	0.31	37.5	0.32	36.0	0.39	36.9	0.34	36.2	0.38
	4000	35.3	0.41	33.9	0.44	37.2	0.39	32.0	0.49	35.7	0.40	33.6	0.48
ALARM	1000	14.8	0.74	13.1	0.78	8.2	0.87	10.8	0.82	7.8	0.88	7.9	0.88
	4000	6.7	0.89	6.2	0.90	5.6	0.92	5.6	0.91	5.1	0.93	5.4	0.92

* The best result for each dataset is highlighted in bold.

i.e.,

$$\Phi(\mathcal{H}) = \cup_{h \in \mathcal{H}} \Phi(h) \quad (11)$$

Proof: Firstly, we assert that the constraint strength of path existence on the DAG structure is not weaker than that of the node order:

$$\Phi(g) \subset \mathcal{O}_{A < B}, \quad \forall g \in \mathcal{G}_{A \rightsquigarrow B} \quad (12)$$

The correctness of (12) relies on the fact that the PEC $A \rightsquigarrow B$ forces A to be a parent or ancestor of B , which requires that the topological ordering of A must be before B . Therefore, the node topological orderings of DAGs satisfying $A \rightsquigarrow B$ must satisfy the OC $A < B$.

Taking the complement of (12), we get:

$$\Phi(g)^c \supset \mathcal{O}_{A < B}^c = \mathcal{O}_{B < A}, \quad \forall g \in \mathcal{G}_{A \rightsquigarrow B} \quad (13)$$

where the superscript c represents the complement of a set. The equality of the above equation holds because A must be either before or after B . Then, taking both sides of (13) as variables for Ψ , we get:

$$\Psi(\mathcal{O}_{B < A}) \subset \Psi(\Phi(g)^c), \quad \forall g \in \mathcal{G}_{A \rightsquigarrow B} \quad (14)$$

The left side of the above equation represents all DAGs that structure satisfy the order $B < A$, which are related to the space $\mathcal{G}_{A \rightsquigarrow B}$ in a complex manner. Going further, it can be observed that:

$$g \notin \Psi(\Phi(g)^c), \quad \forall g \in \mathcal{G} \quad (15)$$

This is because $\Phi(g)^c$ represents the set of node topological orderings that are inconsistent with g . DAGs consistent with these orderings must not include g . Combining (14) and (15), it

can be found:

$$g \notin \Psi(\mathcal{O}_{B < A}), \quad \forall g \in \mathcal{G}_{A \rightsquigarrow B} \quad (16)$$

In this way, a relationship between the PFC and the OC can be obtained:

$$\Psi(\mathcal{O}_{B < A}) \subset \mathcal{G} - \mathcal{G}_{A \rightsquigarrow B} = \mathcal{G}_{A \not\rightsquigarrow B} \quad (17)$$

At this point, half of the proof has been completed. To establish the final conclusion of Lemma 1, we now only need to demonstrate that the reverse of (17) is also valid, i.e.,

$$\mathcal{G}_{A \not\rightsquigarrow B} \subset \Psi(\mathcal{O}_{B < A}) \quad (18)$$

For each $g \in \mathcal{G}_{A \not\rightsquigarrow B}$, following condition holds:

$$g \in \mathcal{G}_{B \rightsquigarrow A} \cup \mathcal{G}_{B \not\rightsquigarrow A} \quad (19)$$

If $g \in \mathcal{G}_{B \rightsquigarrow A}$ holds, then it is evident that $g \in \Psi(\mathcal{O}_{B < A})$. If $g \in \mathcal{G}_{B \not\rightsquigarrow A}$, it suffices to find an order $o \in \mathcal{O}_{B < A}$ such that $g \in \Psi(o)$, which implies that $g \in \Psi(\mathcal{O}_{B < A})$. The proof of this statement is constructive. In fact, the corresponding node order o can be generated for g in an iterative manner:

- *Step 1:* If all root nodes of g do not contain A , place these root nodes in any order after the existing node topological sequence and delete these nodes. If all root nodes of g contain A and the existing node topological sequence does not contain B , place all root nodes except A in any order after the existing node topological sequence and delete these nodes.
- *Step 2:* Repeat step 1 until all nodes of g are deleted.

Since the above node topological ordering generation process ensures that the obtained node topological ordering has B before A , the only aspect to consider is whether this process can

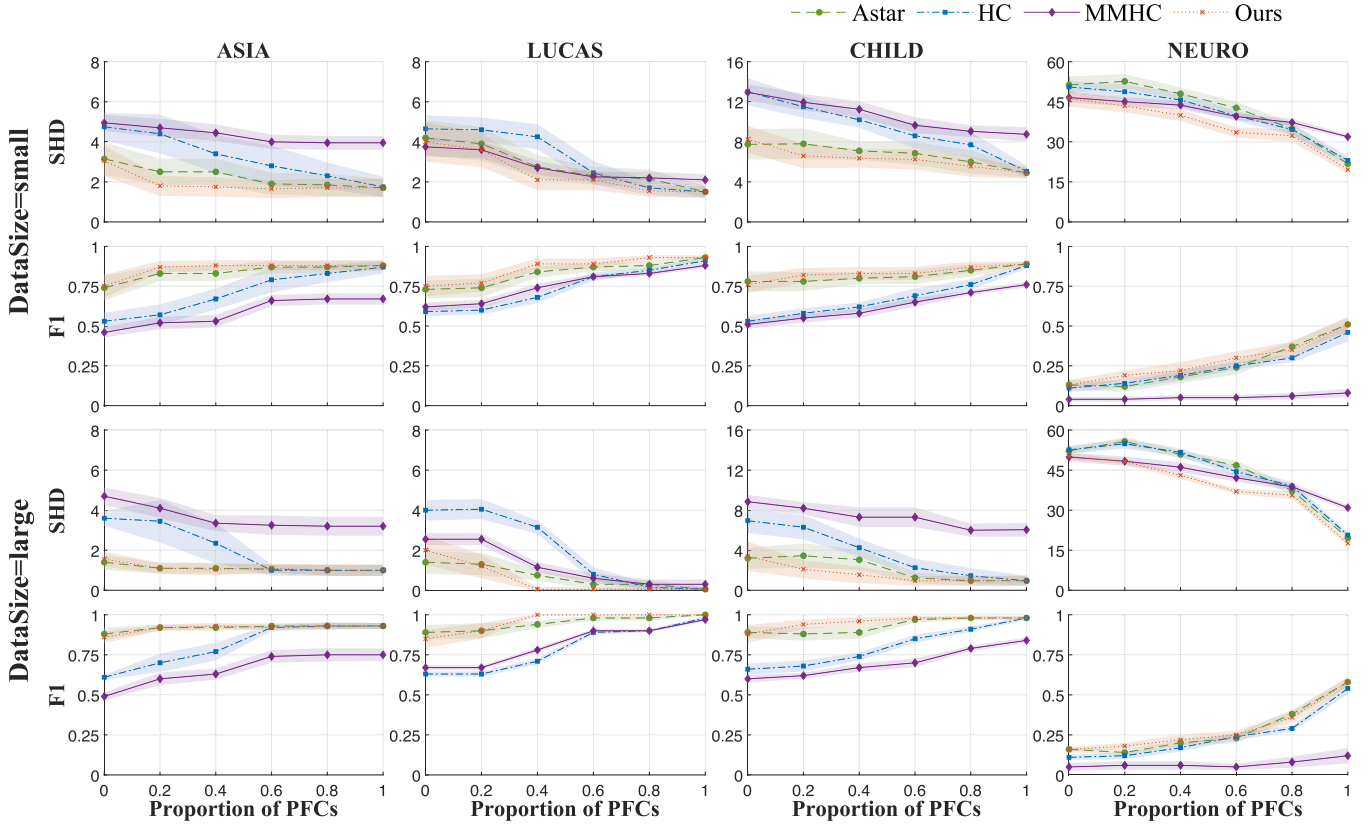


Fig. 2. Performance of different methods under different proportions of PFC.

be completed successfully. In fact, if at some point the root nodes of g contain only A , and the existing node topological sequence does not include B , then A is the ancestor or parent of all remaining nodes (including B naturally), which contradicts $B \not\sim A$. The above discussion demonstrates:

$$g \in \Psi(\mathcal{O}_{B < A}), \quad \forall g \in \mathcal{G}_{A \not\sim B} \quad (20)$$

which implies (18) holds. Combining (17), Lemma 1 holds. $\square$

In summary, Lemma 1 shows that the single PFC $A \not\sim B$ can be replaced with the OC $B < A$, indicating that the optimal DAG consistent with a PFC $A \not\sim B$ is the same as that consistent with the OC $B < A$.

This section then proves Lemma 2, which discusses the difference between OC and the PFC by observing the impact of multiple constraints on the DAG space.

Proof: Based on Lemma 1, we have:

$$\Psi(o_i) \subset \mathcal{G}_{p_i}, \forall o_i \in S(<) \text{ \& } \text{corresponding } p_i \in S(\not\sim) \quad (21)$$

This implies that:

$$\Psi(\cap_{o_i \in S(<)} \mathcal{O}_{o_i}) \subset \cap_{o_i \in S(<)} \Psi(o_i) \subset \cap_{p_i \in S(\not\sim)} \mathcal{G}_{p_i} \quad (22)$$

Thus, Lemma 2 holds. $\square$

Unlike Lemma 1, when using more than one constraint, the DAG spaces satisfying the two constraints may not necessarily be equal. For example, take $S(<) = \{A < B, B > A\}$, it is

clear that:

$$\mathcal{O}_{A < B} \cap \mathcal{O}_{B < A} = \phi \quad (23)$$

where ϕ represents an empty set. Therefore, we can obtain:

$$\Psi(\mathcal{O}_{A < B} \cap \mathcal{O}_{B < A}) = \phi \quad (24)$$

However, for the corresponding PFC $S(\not\sim) = \{A \not\sim B, B \not\sim A\}$, there is:

$$\mathcal{G}_{A \not\sim B} \cap \mathcal{G}_{B \not\sim A} \neq \phi \quad (25)$$

This indicates the existence of some DAGs that satisfy a set of PFCs but do not meet the OCs. Consequently, OCs may exclude more candidate DAGs, leading to a stronger restriction on the DAG space than PFCs.

D. Constraints Quality Assessment of Structural Constraints

This section analyzes the differences in constraint quality among various types of structural constraints. Intuitively, since the absence of a path is a more frequent property in sparse graphs than the existence of a specific direct edge, an LLM making a “forbidden” assertion has a higher statistical probability of being correct even under noisy conditions. Building on this intuition, we establish a lower bound for the quality of structural constraints and derive an approximate ranking of their quality. Notably, under certain assumptions, we also provide proof of the relationship between the constraint quality of PEC and PFC.

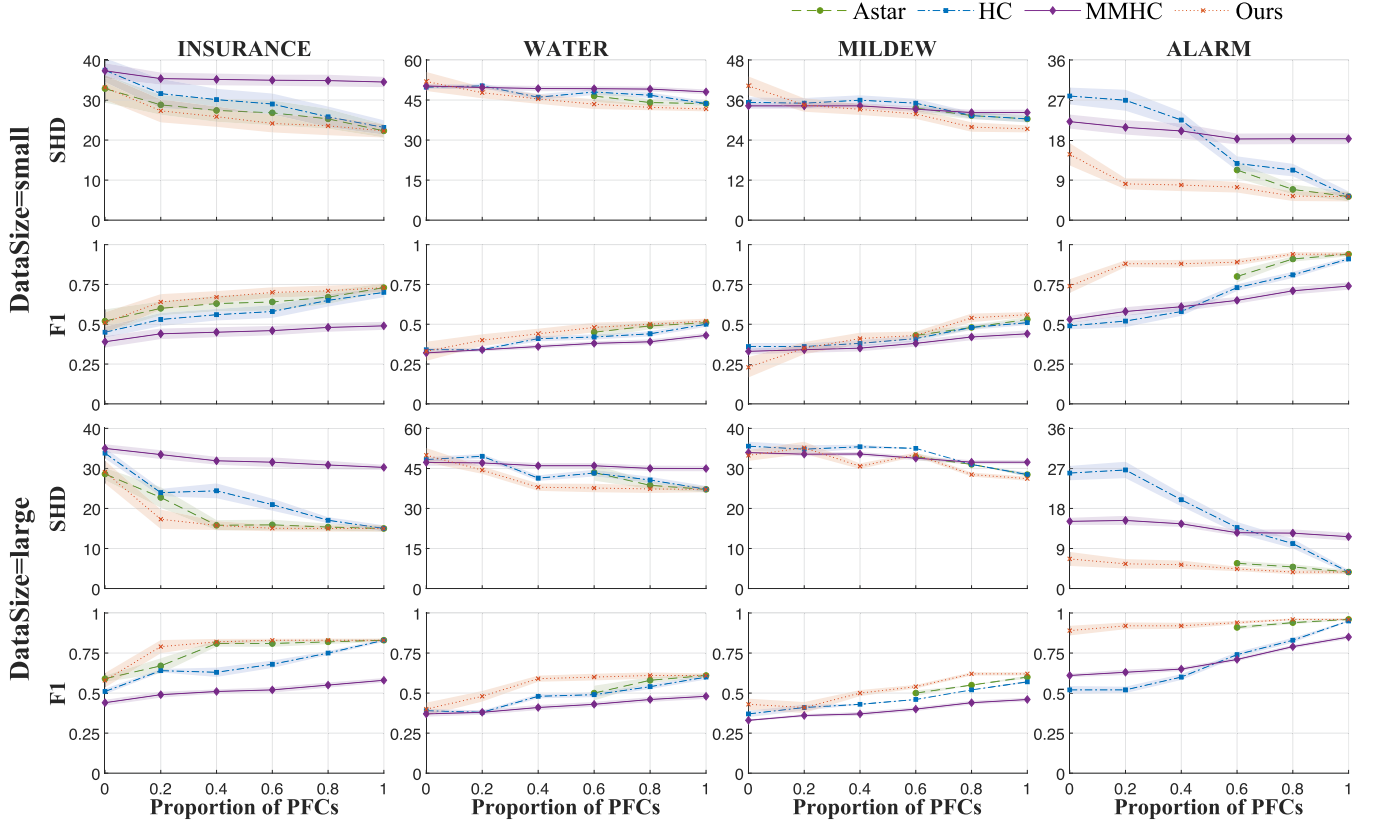


Fig. 3. Performance of different methods under different proportions of PFC. For the WATER, MILDEW, and ALARM, Astar cannot obtain results within 12 hours when the proportion of PFC is low because the search space is too large.

The process for analyzing the lower bound of constraint quality is consistent across all constraints. Here, we use EEC as an example for illustration. Based on Definition 6, The quality of the EEC is:

$$\begin{aligned} CQ(EEC) &= P(M_I(x_i, x_j) = G_{i,j} | M_I(x_i, x_j) = 1) \\ &= P(\hat{G}_{i,j} = 1 | M_I(x_i, x_j) = 1) \end{aligned} \quad (26)$$

where G represents the ground of DAG. According to the law of total probability:

$$\begin{aligned} &P(G_{i,j} = 1) \\ &= P(G_{i,j} = 1 | M_I(x_i, x_j) = 1)P(M_I(x_i, x_j) = 1) \\ &\quad + P(G_{i,j} = 1 | M_I(x_i, x_j) = 0)P(M_I(x_i, x_j) = 0) \end{aligned} \quad (27)$$

and the research findings that LLMs have some ability to give causal relationships (see Section II-A):

$$\begin{aligned} &P(G_{i,j} = 1 | M_I(x_i, x_j) = 1) \\ &> P(G_{i,j} = 1 | M_I(x_i, x_j) = 0) \end{aligned} \quad (28)$$

We can obtain a lower bound on the constraint quality:

$$CQ(EEC) > P(G_{i,j} = 1) \quad (29)$$

This shows that the lower bound of the constraint quality of the EEC is related to the proportion of edges in the graph. Similarly, this conclusion also holds for other types of constraints.

Therefore, for the lower bound of the constraint quality, we have:

$$EEC < PEC < OC < PFC < EFC \quad (30)$$

This chain of relationship can be extended to constraint quality to a certain extent, especially for EECs and PECs. On the one hand, since the number of paths in a DAG is generally much larger than the number of edges, the constraint quality lower bound of the PEC is much larger than that of the EEC. On the other hand, since the performance of current LLMs is still subject to problems such as hallucinations, the real constraint quality, and the quality lower bound are similar. Therefore, it can be considered that the constraint quality of the PEC is greater than that of the EEC. This will also be shown and explained in the experimental section.

Under further assumptions, the constraint quality relationship between PEC and PFC can be derived. Specifically, When I is the instruction to judge the relationship R , this paper further assumes that the results given by LLMs are related to the strength of the relationship R between variables:

Assumption 2: $M_{I_R}(x_i, x_j) = 1_{(R(x_i, x_j) > \theta)}$, where θ is the threshold for determining if the relationship holds, and $R(x_i, x_j)$ represents the strength of the relationship R between variables.

Specifically, when the instruction I requires identifying pairs of variables in a given set of variables that have a causal relationship, LLMs are assumed to output a set:

$$\{(i, j) : M_{I_R}(x_i, x_j) = 1\} = \{(i, j) : R(x_i, x_j) > \theta\} \quad (31)$$

Next, we consider the incorrect output in the prediction of LLMs:

$$\{(i, j) : M_{IR}(x_i, x_j) = 1, M_{IR}(x_i, x_j) \neq \hat{y}_{i,j}\} \quad (32)$$

For PEC and PFC, their capacities can be simplified as:

$$\begin{aligned} \text{error}_{\text{PEC}} &= |\{(i, j) \mid R_{\text{PEC}}(x_i, x_j) > \theta, G_{i,j} = 0\}| \\ &= \sum_{i,j} (1 - G_{i,j}) \mathbf{1}_{R_{\text{PEC}}(x_i, x_j) > \theta} \\ \text{error}_{\text{PFC}} &= |\{(i, j) \mid R_{\text{PFC}}(x_i, x_j) > \theta, G_{i,j} = 1\}| \\ &= \sum_{i,j} G_{i,j} \mathbf{1}_{R_{\text{PFC}}(x_i, x_j) > \theta} \end{aligned} \quad (33)$$

where $R_{\text{PEC}}(x_i, x_j)$, $R_{\text{PFC}}(x_i, x_j)$ denotes the degree of path causality and non-path causality between variables x_i and x_j encoded by the LLM, respectively. This relationship can be expressed as:

$$\begin{aligned} R_{\text{PEC}}(x_i, x_j) &= \mu_{i,j} + \epsilon \\ R_{\text{PFC}}(x_i, x_j) &= \mu'_{i,j} + \epsilon \end{aligned} \quad (34)$$

Here, $\mu_{i,j} \in [0, 1]$ represents the true path causality degree, $\mu'_{i,j} \in [0, 1]$ is the true degree of path non-causality, and ϵ is a random error variable.

We then discuss the properties of $\mu_{i,j}$ and $\mu'_{i,j}$. Since $\mu_{i,j}$ and $\mu'_{i,j}$ have opposite definitions from causality perspective, there is a certain numerical relationship between them:

$$\mu_{i,j} + \mu'_{i,j} = 1 \quad (35)$$

In addition, When there is causality between x_i and x_j , the greater the certainty that x_i is the cause of x_j , the greater the certainty that x_j is not the cause of x_i . Therefore, when there is a path from x_i to x_j in G , we consider:

$$\mu_{i,j} > 0.5, \mu_{i,j} + \mu_{j,i} = 1 \quad (36)$$

Conversely, when x_i and x_j are not connected in G , we have:

$$\mu_{i,j} < 0.5, \mu_{j,i} < 0.5 \quad (37)$$

This corresponds to the case where x_i and x_j are two unrelated variables.

Substituting (34), (35), (36) and (37) into (33), we can obtain:

$$\begin{aligned} \text{error}_{\text{PEC}} &= \sum_{i,j} (1 - G_{i,j}) \mathbf{1}_{\mu_{i,j} + \epsilon > \theta} > \sum_{i,j} G_{j,i} \mathbf{1}_{1 - \mu'_{i,j} + \epsilon > \theta} \\ &> \sum_{i,j} G_{j,i} \mathbf{1}_{\mu'_{j,i} + \epsilon > \theta} = \text{error}_{\text{PFC}} \end{aligned} \quad (38)$$

This indicates that the PEC given by LLMs has more errors. And for the constraint quality:

$$CQ(\text{PEC}) = 1 - \frac{\text{error}_{\text{PEC}}}{|S(\rightsquigarrow)|} < 1 - \frac{\text{error}_{\text{PFC}}}{|S(\nearrow)|} = CQ(\text{PFC}) \quad (39)$$

Here, $S(\rightsquigarrow) = \{(i, j) : M_{\text{PEC}}(x_i, x_j) = 1\}$ and $S(\nearrow) = \{(i, j) : M_{\text{PFC}}(x_i, x_j) = 1\}$. The reason why the inequality holds is, firstly, due to (38), and secondly, because the PFC given by LLM is more than PEC, which will also be shown in the experiment. In summary, the result indicates that the PFC obtained using LLM has higher constraint quality than PEC.

E. Analysis of Structural Constraint Suitability

Based on the constraint strength and constraint quality of five kinds of constraints, this section evaluates their suitability for LLM-based CSL

First, the EEC has a low constraint quality, and using it for CSL may introduce a lot of errors. Therefore, the EEC is not suitable as a target for obtaining priors using LLMs. The EFC has the highest constraint quality and can be used to learn causal structures. However, the constraint strength of EFC is low and their effect is limited. In fact, for a common sparse graph, most of the node pairs have no direct causal relationship, and a large number of node pairs themselves will not be learned by the CSL algorithm. Therefore, although the quality of the EFCs is high, most of the EFCs are meaningless, and whether they are used depends on the specific situation.

We focus on the other three structural constraints. Theoretically, the constraint strength and constraint quality of PEC and PFC are both at a certain level and not too low. Therefore, when using priors, PEC and PFC may be considered. Finally, since PEC corresponds to an order constraint, it is natural to integrate an order constraint when adopting a PEC.

V. CAUSAL STRUCTURE LEARNING WITH JOINT STRUCTURAL CONSTRAINTS

Assuming that PECs, PFCs, EFCs, and OCs have already been acquired, this section describes how to construct a CSL framework that integrates these four types of prior knowledge. The objective is to identify a DAG that fits the observed data while satisfying the given structural constraints. The search backbone adopted here is related to conventional order-based structure learning, which is retained for computational efficiency. Our focus, however, is not on proposing a fundamentally new search paradigm, but on developing a unified mechanism for jointly incorporating heterogeneous constraints induced by LLM priors. Unlike settings that consider only limited forms of prior knowledge, our framework must simultaneously accommodate both global and local constraints, which operate at different structural levels.

To this end, we employ a two-level optimization search strategy. At the first level, the DAG space is partitioned according to node orders so that global constraints can be efficiently represented and enforced. At the second level, local searches are conducted within the resulting subspaces while further incorporating edge- and path-related constraints such as PECs, PFCs, and EFCs. This organization is designed to support the coordinated integration of global and local constraints within a single search process, and thus serves as the optimization component of our reliability-driven framework for handling uncertain LLM-derived knowledge.

Specifically, the first step of the search strategy is to divide the DAG space into multiple subspaces according to node order:

$$\mathcal{G} = \bigcup_{o \in \mathcal{P}} \Psi(\mathcal{O}_o) \quad (40)$$

Here, $\mathcal{P}$ represents all $n!$ permutations of n nodes, and $\mathcal{O}_o$ is the DAG subspace that conforms to permutation o . The goal of the first-level search is to find the optimal node order and the

TABLE V
CSL RESULTS BASED ON REAL CONSTRAINTS GIVEN BY LLM

Network	DataSize	Constraint	Extra	Missing	Reverse	Precision	Recall	F1	SHD	SID
ASIA	250	No Constraint	0.65	1.80	0.60	0.81	0.70	75.12±14.43	3.05±1.54	10.65
		OC & EFC	0.20	1.40	0.00	0.97	0.82	89.04 ±6.97	1.60 ±0.99	3.05
		Major Voted	0.25	1.45	0.00	0.96	0.82	88.42±6.48	1.70±0.92	3.10
	1000	No Constraint	0.10	0.95	0.50	0.92	0.82	86.46±7.50	1.55±0.76	4.50
		OC & EFC	0.10	0.95	0.00	0.99	0.88	93.04 ±3.97	1.05 ±0.60	1.10
		Major Voted	0.10	0.95	0.00	0.99	0.88	93.04 ±3.97	1.05 ±0.60	1.10
LUCAS	250	No Constraint	0.70	1.45	1.85	0.78	0.72	74.88±13.05	4.00±2.00	25.25
		OC & EFC	0.55	1.40	1.70	0.80	0.74	76.97±11.94	3.65±1.87	23.30
		Major Voted	0.25	1.25	0.65	0.92	0.84	87.77 ±5.09	2.15 ±0.93	13.05
	1000	No Constraint	0.25	0.05	1.70	0.84	0.85	84.79±11.46	2.00±1.49	13.55
		OC & EFC	0.20	0.05	1.25	0.88	0.89	88.63±9.76	1.50±1.19	10.35
		Major Voted	0.10	0.05	0.50	0.95	0.95	95.23 ±4.29	0.65 ±0.59	4.45
CHILD	500	No Constraint	0.50	4.95	2.80	0.84	0.69	75.71±5.49	8.25±1.37	115.00
		OC & EFC	0.45	5.10	2.00	0.88	0.72	78.91±5.08	7.55±1.36	105.05
		Major Voted	0.20	4.95	0.30	0.98	0.79	87.28 ±3.06	5.45 ±1.05	63.60
	2000	No Constraint	0.05	1.00	2.35	0.90	0.87	88.27±5.64	3.40±1.47	59.25
		OC & EFC	0.05	1.00	1.95	0.92	0.88	89.90±4.33	3.00±1.17	52.55
		Major Voted	0.05	1.00	0.05	1.00	0.96	97.65 ±2.13	1.10 ±0.85	15.15
NEURO	500	No Constraint	22.25	27.65	1.45	0.14	0.12	13.05±3.09	51.35±1.69	582.70
		OC & EFC	19.60	27.65	1.45	0.15	0.12	14.04±4.16	48.70±1.32	588.90
		Major Voted	22.60	25.00	0.85	0.14	0.13	14.08 ±4.35	48.45 ±1.32	569.70
	2000	No Constraint	24.15	26.85	1.10	0.17	0.15	16.88±1.87	52.20±0.97	579.85
		OC & EFC	22.85	26.85	1.55	0.16	0.14	15.80±2.67	51.25±0.85	572.15
		Major Voted	24.00	24.85	1.05	0.17	0.17	17.69 ±2.45	49.90 ±0.64	564.95
INSURANCE	500	No Constraint	2.05	24.75	6.35	0.71	0.40	51.36±6.50	33.15±3.07	494.20
		OC & EFC	2.35	24.80	4.65	0.76	0.43	55.24±5.99	31.80±2.98	479.70
		Major Voted	2.00	23.75	0.15	0.93	0.54	68.29 ±3.32	25.90 ±2.27	407.80
	2000	No Constraint	3.10	17.25	8.75	0.69	0.50	57.89±4.45	29.10±2.83	398.75
		OC & EFC	2.35	16.95	5.50	0.79	0.57	66.06±9.39	24.80±5.43	350.95
		Major Voted	1.00	15.85	0.00	0.97	0.70	81.09 ±1.41	16.85 ±1.18	243.65
WATER	1000	No Constraint	2.40	43.45	8.10	0.58	0.22	31.79±5.99	53.95±3.56	474.60
		OC & EFC	0.10	43.60	0.00	1.00	0.34	50.61 ±1.50	43.70 ±0.92	364.45
		Major Voted	0.20	43.65	0.00	0.99	0.34	50.47±1.61	43.85±1.04	365.30
	4000	No Constraint	3.10	38.30	8.55	0.62	0.29	39.55±4.00	49.95±2.67	488.80
		OC & EFC	0.20	36.80	0.00	0.99	0.44	61.21 ±1.22	37.00 ±0.97	347.60
		Major Voted	0.20	36.80	0.00	0.99	0.44	61.21 ±1.34	37.00 ±1.08	347.60
MILDEW	1000	No Constraint	4.70	26.60	11.95	0.31	0.16	21.26±6.35	43.25±2.81	683.45
		OC & EFC	2.70	29.50	0.05	0.86	0.36	50.45 ±2.64	32.25 ±1.68	416.65
		Major Voted	2.10	28.90	2.40	0.77	0.32	45.09±3.05	33.40±1.60	472.40
	4000	No Constraint	4.95	21.05	9.25	0.53	0.34	41.37±3.53	35.25±1.37	637.65
		OC & EFC	4.45	24.70	0.00	0.83	0.46	59.37 ±1.22	29.15 ±0.88	375.80
		Major Voted	4.95	25.00	2.65	0.71	0.40	51.02±3.27	32.60±1.98	453.60
ALARM	1000	No Constraint	1.95	4.90	7.95	0.77	0.72	74.46±4.26	14.80±2.50	197.45
		OC & EFC	1.75	5.10	5.35	0.83	0.77	80.20 ±4.74	12.20 ±2.63	155.85
		Major Voted	1.25	4.70	6.45	0.82	0.76	78.73±3.95	12.40±2.41	162.20
	4000	No Constraint	0.40	3.00	3.25	0.92	0.86	88.94±3.02	6.65±1.57	85.90
		OC & EFC	0.65	3.10	2.60	0.93	0.88	90.02 ±4.02	6.35 ±2.11	77.65
		Major Voted	0.35	3.00	3.40	0.91	0.86	88.65±3.45	6.75±1.74	87.60

* The better results are in bold between the results using only data and the results using structural constraints.

highest-scoring DAG under that order. The proposed method employs a greedy strategy to search across this order space, continually exploring better orders.

At each step, the greedy strategy considers the neighborhood structure of the order. Neighborhood definitions include insertion neighborhoods and swap neighborhoods. If o' is generated by selecting a node from o and inserting it into a new position,

it belongs to an insertion neighborhood; if o' is generated by swapping two adjacent nodes in o , it belongs to a swap neighborhood. This definition aligns with our intuitive understanding of the arrangement of items. The evaluation metric for the search is the highest score of DAGs in $\Phi(O)$, and the detailed search process will be further explained in the second-level optimization. An advantage of this evaluation metric is that it

Algorithm 1: First-level Optimization Search.

```

1: Input: Order constraint set OC, Path existence constraint set
   PEC
2: Output: Optimal DAG
3: Partition DAG space based on node order  $\mathcal{G} = \cup_{o \in \mathcal{P}} \Psi(\mathcal{O}_o)$ 
4: for each  $x_i \rightsquigarrow x_j \in \text{PEC}$  do
5:    $\text{OC} \leftarrow \text{OC} \cup \{x_i < x_j\}$ 
6: end for
7: Initialize a random order  $o$  consistent with OC
8: Find the best DAG  $G \in \Psi(\mathcal{O}_o)$  using Algorithm 2
9:  $o_{\text{Cand}}, G_{\text{Cand}} \leftarrow o, G$ 
10: repeat
11:   for each order  $o'$  in the neighborhood of  $o$  and consistent
       with OC do
12:     Find the optimal  $G' \in \Psi(\mathcal{O}_{o'})$  using Algorithm 2
13:     if  $G'$  is better than  $G_{\text{Cand}}$  then
14:        $o_{\text{Cand}}, G_{\text{Cand}} \leftarrow o', G'$ 
15:     end if
16:   end for
17:   if  $G_{\text{Cand}}$  is better than  $G$  then
18:      $o, G \leftarrow o_{\text{Cand}}, G_{\text{Cand}}$ 
19:   end if
20: until convergence
21: return Optimal DAG  $G$ 

```

requires no additional computation, allowing us to obtain the highest-scoring DAG as the final result while searching for the optimal order. The complete first-level optimization process is detailed in Algorithm 1.

Through the set of OCs, the size of the permutation set $\mathcal{P}$ can be effectively reduced, thereby narrowing down the neighborhood of each order $\mathcal{O}_o$. The bipartiteness of the OC means that each added constraint approximately halves the search space. It is noteworthy that since PECs are stronger than OCs, they can also be incorporated into the set of OCs during the first level of optimization. For each PEC $x_i \rightsquigarrow x_j$, we specify an OC $x_i < x_j$, which prunes any order where x_j precedes x_i . This also illustrates the necessity of optimizing the search at the first level. As long as there are OCs or PECs, the DAG space can be partitioned to optimize the search process.

In the second level of optimization, the proposed method searches for the optimal DAG structure within the subspace $\Phi(\mathcal{O}_o)$ defined by a given order o that simultaneously satisfies PEC, PFC, and EFC. Here, we continue to use the greedy strategy to optimize the DAG. At each step, the greedy strategy explores the neighborhood of the current DAG, including new DAGs generated by adding, deleting, or reversing edges.

To find an effective DAG neighborhood while considering the constraints PEC, PFC, and EFC, we define an initial candidate parent set for each variable, composed of variables that precede it in order o . We then prune these candidate sets according to PFC and PEC: for any $x_i \not\rightsquigarrow x_j \in \text{PFC}$ and $x_i \not\rightsquigarrow x_j \in \text{EFC}$, we remove x_i from x_j 's candidate parent set; for any $x_i \rightsquigarrow x_j \in \text{PEC}$, we remove x_j from x_i 's candidate parent set. This approach takes advantage of the typically stronger nature of PFCs compared to EFCs.

Algorithm 2: Second-level Optimization Search.

```

1: Input: Path existence prior set PEC, Path forbidden set
   PFC, order  $o$ , Edge forbidden set EFC, Dataset  $D$ 
2: Output: Optimal DAG in  $\Psi(\mathcal{O}_o)$ 
3: for each  $x_i \not\rightsquigarrow x_j \in \text{PFC}$  and  $x_i \not\rightsquigarrow x_j \in \text{EFC}$  do
4:   Remove  $x_i$  from candidate parent sets of  $x_j$ 
5: end for
6: for each  $x_i \rightsquigarrow x_j \in \text{PEC}$  do
7:   Remove  $x_j$  from candidate parent sets of  $x_i$ 
8: end for
9: Initialize  $G$  with a random DAG in  $\Psi(\mathcal{O}_o)$ 
10: repeat
11:   for each DAG  $G'$  in the neighborhood of  $G$  do
12:     if  $\psi_e(G') + \psi_f(G') > \psi_e(G) + \psi_f(G)$  or
        ( $\sigma(G') > \sigma(G)$  and
         $\psi_e(G') + \psi_f(G') = \psi_e(G) + \psi_f(G)$ ) then
13:        $G \leftarrow G'$ 
14:     end if
15:   end for
16: until convergence
17: return Optimal DAG  $G$ 

```

Since path constraints are global, merely limiting the candidate parent sets does not guarantee that the final DAG fully complies with all constraints. Therefore, we further define two metrics, $\psi_e(G)$ and $\psi_f(G)$, which represent the number of constraints from PEC and PFC that a DAG G satisfies, respectively. During each iteration of the greedy search, we select a DAG G' from the neighborhood of DAG G only if it meets one of the following conditions:

$$\begin{aligned}
1. & \psi_e(G') + \psi_f(G') > \psi_e(G) + \psi_f(G) \\
2. & \sigma(G') > \sigma(G), \psi_e(G') + \psi_f(G') = \psi_e(G) + \psi_f(G) \quad (41)
\end{aligned}$$

That is, we prioritize DAGs that satisfy more path constraints, and secondarily, among those with the same number of satisfied path constraints, we choose the DAG with the higher score. The complete process is detailed in Algorithm 2.

Note that, if inconsistencies occur within the OC set, the framework can still maintain a reachable first-level search space through a simple conflict-handling mechanism. In the case of a direct pairwise contradiction, such as the simultaneous presence of $x < y$ and $y < x$, these two constraints are treated as mutually cancelling order evidence and removed from the initial OC set. After this preprocessing step, the algorithm does not enforce any order preference for that variable pair.

For more general cases in which the remaining OC set is still globally inconsistent, the first-level search can be relaxed into a soft-constraint optimization problem. Specifically, the search objective can be augmented with an order-violation metric that counts how many OCs are violated by a candidate node ordering. The algorithm then prioritizes orderings with fewer violated OCs, rather than requiring all OCs to be satisfied exactly. This treatment is analogous to the second-level optimization over path constraints and allows the framework to remain functional under noisy prior acquisition.

VI. EXPERIMENTAL SETUP

Methods and Datasets Used: 8 causal networks, comprising ASIA, LUCAS, CHILD, NEURO, INSURANCE, WATER, MILDEW and ALARM are utilized to evaluate the structural constraints studied in this paper and the proposed framework, where LUCAS is derived from the causality workbench project², NEURO is obtained from both domain experts and known medical literature [51], and other networks are obtained from the Bayesian Network Repository³. Table II summarizes these benchmark networks, including their sizes and application domains, to provide a diverse set of benchmark scales and causal application scenarios.

In evaluating the performance of the structure constraint, we select several existing baseline methods in addition to the proposed approach, including Hill Climbing (HC) [55], MMHC [56], and Astar [57]. These methods tackle the CSL problem from different angles. HC is a commonly used score-based method, MMHC is a hybrid approach, and Astar is an exact search method. HC, MMHC, and the proposed method employ BIC score and require no parameters.

Metrics: There are two main indicators used in this paper: Structural Hamming Distance (SHD) and F1. The definition of SHD is as follows:

Definition 7: Let $\mathcal{G}$ be the space of DAGs over n variables. The Structural Hamming Distance is defined as:

$$SHD : \mathcal{G} \times \mathcal{G} \rightarrow \mathbb{N}$$

$$(g_1, g_2) \rightarrow \#\{(v_i, v_j) \in V^2 \mid g_1 \text{ and } g_2 \text{ do not have the same type of edge between } v_i \text{ and } v_j\}$$

where $\mathbb{N}$ represents the set of natural numbers.

SHD quantitatively assesses the disparity in edge structures between a learned DAG and the ground truth. SHD comprises three distinct components: extra edges (those present in the learned DAG but absent in the ground truth), missing edges (those in the ground truth but omitted in the learned DAG), and reverse edges (edges that exist in opposing directions between the ground truth and the learned DAG).

Experimental Hardware: All experiments are run on a single core of AMD Ryzen9 7950x CPU @ 4.5 GHz with a memory limit of 32 GB and a GeForce RTX 3090 GPU with 24 GB graphics memory.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

To prove the correctness of the conclusions mentioned above and the effectiveness of the proposed framework, this paper conducts four experiments. The first experiment shows the constraint quality of the structural constraints given by LLM to prove the assertion mentioned in Section IV. The second experiment provides a basic verification of the performance of PFC. The third experiment shows the comprehensive improvement of CSL by different constraints under the proposed framework. Finally, we conduct tests in real scenarios.

A. Verification of the Constraint Quality of the Structural Constraints Given by LLM

This experiment evaluates the performance of various structural constraints provided by the LLM. Specifically, we use GPT-4o as the test object, prompting it for each causal network and then post-processing its text output to derive the corresponding structural constraints. The complete set of prompt templates and the detailed parsing logic for post-processing are documented in Supplementary material A, online available. The statistical results are presented in Table III.

We begin by focusing on precision, as it reflects the quality of the structural constraints. Notably, EFC and OC exhibit the highest constraint quality, with average precision of 0.97 and 0.90 respectively, demonstrating the LLM's strong capability in determining both causal and non-causal orders. PFC has an average precision of 0.87, which, while slightly lower, still represents a viable structural constraint.

Recall provides insights into the efficiency of the acquisition process. The observed trade-off between recall and CQ suggests that acquiring more constraints often comes at the cost of lower precision. In particular, despite its high recall, EEC shows the lowest constraint quality, with precision often falling below 0.5. This suggests that relying on EEC as a prior does not necessarily improve result precision and may even be detrimental, which is consistent with our earlier assertions. Meanwhile, PEC achieves a slightly higher constraint quality than EEC, making it somewhat usable in specific contexts, such as when variable scenarios or semantics are relatively straightforward (e.g., the ASIA network).

Overall, OC, PFC, and EFC demonstrate high constraint quality, which is consistent with the theoretical analysis in this paper. The balanced F1 scores for these constraints further confirm their effectiveness as reliable priors. On certain causal networks, their precision exceeds 0.9, indicating that these three constraints are comparatively reliable choices for LLM-based CSL.

To further evaluate the robustness of our framework across different model architectures, we extended the constraint extraction experiments to include Gemini 2.5, GPT-4.1, Claude 4.5 Haiku Thinking, and DeepSeek v3.2.⁴ The comparative analysis reveals that while the absolute precision and recall vary slightly between models, the relative quality ranking of the five constraint types remains broadly consistent. The detailed multi-model results are reported in Supplementary material B, online available.

B. Performance Improvement of Path Forbidden Constraints on Different CSL Methods

In the previous section, we demonstrated the credibility of PFC through empirical experiments. Since PFC is rarely selected to assist CSL in existing studies, we provide basic validation for the improvement of CSL performance with PFC in this experiment. For each causal network, we generate datasets of two different sizes to simulate small-scale and large-scale datasets, where the specific data size is determined by the

²<https://www.causality.inf.ethz.ch/>

³<https://www.bnlearn.com/bnrepository/>

⁴The specific version numbers are Gemini-2.5-flash-thinking, GPT-4.1-2025-04-14, Claude-haiku-4.5-20251001-thinking, Deepseek-v3.2-exp

number of nodes in the network. For each causal network and each dataset size, we randomly generated the dataset 20 times and obtained the mean and variance. The results are shown in Figs. 2 and 3. The results of different proportions of PFCs across various datasets are demonstrated, where these constraints are randomly generated in proportion from the non-path pairs set of each dataset's variables. Four methods, including HC, MMHC, Astar, and the proposed method in this paper, are utilized. Note that the HC and MMHC methods treat PFCs as EFCs, due to the design of these methods not supporting path existence or forbidden constraints at the global level.

It is evident that as the proportion of PFC increases, the F1 scores of various methods on each dataset show a gradual upward trend, while SHD generally decreases, indicating the effectiveness of PFC. Among them, the framework proposed in this paper has the best performance overall.

However, in some cases (such as with the WATER and MILDEW datasets), there is a performance limit to the improvements that PFC can provide for CSL, primarily due to the difficulty in identifying certain causal relationships. As a prior, PFC is effective in eliminating redundant causal relationships in the DAG, but cannot easily help discover previously unidentified causal connections. Consequently, even with complete knowledge of PFCs, CSL results may still diverge from the true causal structure. This observation similarly applies to the EEC, EFC, and PEC.

In such situations, EEC or PEC may be needed to enforce causal constraints on the CSL algorithm, thereby enhancing result accuracy. Conversely, when the CSL method has already identified most or even redundant causal relationships, PFC or EFC can be useful to refine and correct those results.

C. Performance Analysis of Path Existence and Path Forbidden Joint Constraints

As discussed in the previous section, structural constraints may encounter performance limits in certain scenarios. Therefore, combining multiple constraints is essential to achieve better outcomes. The proposed CSL framework is designed to integrate multiple structural constraints, fully leveraging their information to ensure robust and effective use of constraints.

In particular, this section examines the combined effect of using PFC and PEC within CSL, with the results presented in Table IV. For each dataset, we conduct 20 experiments. We added 0, 20%, or 40% PFC and/or PEC during the CSL process and grouped the results based on the sum of the proportions of added priors. The optimal SHD and F1 corresponding to each data are marked in bold. It can be seen that, generally, the more priors that are added, the better the performance of CSL. Within the same group, different PEC/PFC proportions yield varying results across datasets, indicating that different datasets have distinct requirements for structural constraints. For instance, in the MILDEW and WATER datasets, PEC provides greater improvements, while for NEURO and ALARM, the combination of PFC and PEC yields the best results.

D. Reliable and Effective Autonomous Causal Discovery in Real-World Scenarios

The previous experiments utilized simulated constraints to establish a performance baseline. This section further evaluates whether the proposed method can serve as a reliable autonomous discovery pipeline in realistic settings where the correctness of priors cannot be verified in advance.

As a semi-automated reference setting, we first evaluated the potential benefit of high-quality LLM-derived priors. For each dataset, we identified structural constraints with a precision exceeding 0.90 and randomly sampled a 20% subset for each experimental run. To account for sampling variability and ensure the robustness of our findings under different prior combinations, each result presented in Table V represents the mean and standard deviation across 20 independent trials, with each trial using a different randomly sampled subset. Within these evaluations, we also report the Structural Intervention Distance (SID), which evaluates the discrepancy between the interventional distributions implied by the learned graph and the ground truth, thus providing a causality-oriented measure of structural consistency. The results indicate that such constraints can improve CSL performance, particularly in the WATER and MILDEW datasets. This stability across multiple random samples highlights the potential of reliably filtered LLM-derived priors to support causal learning. Note that, for simpler networks like ASIA, where data-driven results are already robust, the improvement is marginal due to the inherent saturation of structural information.

To evaluate automation under realistic deployment conditions, we further implemented a fully automatic filtering strategy that does not rely on any ground-truth knowledge. Rather than selecting constraints based on known precision, we first assessed the five types of structural constraints according to their average cross-model consensus over candidate relations, and then retained the constraint types with relatively stable agreement (i.e., those for which the proportion of model-supported candidate relations exceeded a threshold of 0.7 when aggregated across all candidate edges and models). Within the selected types, the final constraints were determined by majority voting across models.

In practice, constraint types such as PFC and OC tend to exhibit more stable inter-model agreement, whereas EEC often shows substantial disagreement. This observation is consistent with our theoretical analysis of prior-type compatibility and helps explain why consensus-based filtering is more effective for some constraint categories than for others. As shown in Table V, the fully automated pipeline remains competitive with, and often improves upon, the data-only baseline, suggesting that the proposed framework is promising for expert-scarce environments where ground-truth filtering is unavailable.

VIII. DISCUSSION

a) Scope and Comparative Positioning: The proposed framework complements existing work on LLMs in causality by focusing on the reliability and utility of structural constraint types for causal structure learning. Previous studies such as Kiciman et al. [46] and Romanou et al. [48] mainly evaluate

the causal reasoning or semantic capabilities of LLMs, whereas our work studies how different forms of LLM-derived structural priors behave once they are integrated into CSL. Related methods such as Takayama et al. [40] and Zanna et al. [58] also use LLMs to support discovery, but their use of constraints is primarily motivated empirically rather than through an explicit analysis of structural risk and pruning effects. In contrast, our framework introduces constraint quality and constraint strength as two analytical dimensions for evaluating which prior types are more suitable for reliable integration under noisy acquisition.

b) Risks and Hallucinations: A reduction in search space does not necessarily imply genuine discovery gains. In CSL, aggressive pruning can improve efficiency, but if the retained priors are incorrect, it may also exclude the ground-truth structure from the feasible set. This is particularly important under LLM-derived priors, where hallucinations may not only introduce incorrect edges but also propagate by changing which additional edges remain admissible. For example, the acyclicity constraint in CSL can make some correct edges effectively unlearnable once an incorrect edge has already been enforced. Compared with strict existence assertions, global forbidden constraints tend to act more softly by reducing the candidate space without directly forcing the inclusion of potentially erroneous structures.

c) Benchmark Data Necessity: This research uses both real-world datasets and simulation-based benchmarks to evaluate the proposed framework. Simulation-based benchmarks remain important in the current research landscape because they provide controlled ground-truth structures, which makes it possible to quantify structural risk, compare different types of priors, and measure the downstream effects of noisy knowledge in a reproducible way. At the same time, such benchmarks do not fully capture the semantic ambiguity and domain shift encountered in real applications. Therefore, our conclusions should be interpreted as a framework-level analysis of reliability under controlled conditions, rather than as a complete substitute for real-world expert validation. More broadly, the proposed method remains limited by the structural characteristics of the evaluated networks and by the reasoning capabilities of current LLMs.

IX. CONCLUSION

In this paper, we have developed a formal framework for defining and evaluating the strength and quality of constraints derived from LLMs for use in CSL. Our analysis reveals that the commonly employed structural constraints vary significantly in their reliability and their impact on the overall effectiveness of CSL methods. We also introduce a novel CSL framework that integrates multiple structural constraints, thereby enhancing the adaptability of CSL to different LLM outputs and improving both the reliability and robustness of causal inference.

Our experimental evaluations demonstrate that high-quality constraints, particularly path forbidden constraints, can enhance the accuracy of CSL by effectively pruning the DAG space without compromising accuracy. These results not only validate our theoretical framework but also highlight the potential of utilizing LLM-derived constraints in CSL, especially in scenarios where expert knowledge is scarce.

Moving forward, we aim to further investigate the precision of LLM outputs and refine our CSL methods to improve their practical application across various domains. By continuing to refine these approaches, we strive to fully leverage the capabilities of LLMs in automating and enhancing the reliability of causal discovery processes.

REFERENCES

- [1] A. C. Constantinou, Z. Guo, and N. K. Kitson, "The impact of prior knowledge on causal structure learning," *Knowl. Inf. Syst.*, vol. 65, no. 8, pp. 3385–3434, 2023.
- [2] A. C. Constantinou, Y. Liu, K. Chobtham, Z. Guo, and N. K. Kitson, "Large-scale empirical validation of Bayesian network structure learning algorithms with noisy data," *Int. J. Approx. Reasoning*, vol. 131, pp. 151–188, 2021.
- [3] H. Amirkhani, M. Rahmati, P. J. Lucas, and A. Hommersom, "Exploiting experts' knowledge for structure learning of Bayesian networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 11, pp. 2154–2170, Nov. 2017.
- [4] U. Hasan and M. O. Gani, "KCRL: A prior knowledge based causal discovery framework with reinforcement learning," in *Proc. Mach. Learn. Healthcare Conf.*, 2022, pp. 691–714.
- [5] J. Peters, D. Janzing, and B. Schölkopf, "Causal inference on discrete data using additive noise models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2436–2450, Dec. 2011.
- [6] Q. Ye, A. A. Amini, and Q. Zhou, "Federated learning of generalized linear causal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6623–6636, Oct. 2024.
- [7] A. Tank, I. Covert, N. Foti, A. Shojai, and E. B. Fox, "Neural granger causality," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4267–4279, Aug. 2021.
- [8] H. Zhang, Y. Ren, Y. Xia, S. Zhou, and J. Guan, "Towards effective causal partitioning by edge cutting of adjoint graph," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10259–10271, Dec. 2024.
- [9] C. Heinze-Deml, M. H. Maathuis, and N. Meinshausen, "Causal structure learning," *Annu. Rev. Statist. Appl.*, vol. 5, no. 1, pp. 371–391, 2018.
- [10] P. Larranaga, M. Poza, Y. Yurramendi, R. H. Murga, and C. M. H. Kuijpers, "Structure learning of Bayesian networks by genetic algorithms: A performance analysis of control parameters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 9, pp. 912–926, Sep. 1996.
- [11] D. Oyen, B. Anderson, and C. Anderson-Cook, "Bayesian networks with prior knowledge for malware phylogenetics," in *Proc. Workshops 30th AAAI Conf. Artif. Intell.*, 2016, pp. 185–192.
- [12] M. J. Flores, A. E. Nicholson, A. Brunsell, K. B. Korb, and S. Mascaro, "Incorporating expert knowledge when learning Bayesian network structure: A medical case study," *Artif. Intell. Med.*, vol. 53, no. 3, pp. 181–204, 2011.
- [13] M. B. Sesen, A. E. Nicholson, R. Banares-Alcantara, T. Kadir, and M. Brady, "Bayesian networks for clinical decision support in lung cancer care," *PLoS One*, vol. 8, no. 12, 2013, Art. no. e82349.
- [14] E. Y.-J. Chen, Y. Shen, A. Choi, and A. Darwiche, "Learning Bayesian networks with ancestral constraints," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, vol. 29, pp. 2325–2333.
- [15] C. J. Oates, J. Kasza, J. A. Simpson, and A. B. Forbes, "Repair of partly misspecified causal diagrams," *Epidemiology*, vol. 28, no. 4, pp. 548–552, 2017.
- [16] T. Ban, L. Chen, D. Lyu, X. Wang, and H. Chen, "Causal structure learning supervised by large language model," 2023, *arXiv:2311.11689*.
- [17] S. Long, T. Schuster, and A. Piché, "Can large language models build causal graphs?," 2023, *arXiv:2303.05279*.
- [18] Y. Tang, Z. Pan, X. Hu, W. Pedrycz, and R. Chen, "Knowledge-induced multiple kernel fuzzy clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14838–14855, Dec. 2023.
- [19] S. Ji, S. Pan, E. Cambria, P. Martinen, and S. Y. Philip, "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 494–514, Feb. 2021.
- [20] B. Lin et al., "Correctable landmark discovery via large models for vision-language navigation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 8534–8548, Dec. 2024.
- [21] L. Siyao et al., "Bailando : 3D dance GPT with choreographic memory," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14192–14207, Dec. 2023.

- [22] E. De Santis, A. Martino, and A. Rizzi, "Human versus machine intelligence: Assessing natural language generation models through complex systems theory," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 7, pp. 4812–4829, Jul. 2024.
- [23] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.
- [24] K. Choi, C. Cundy, S. Srivastava, and S. Ermon, "LMPriors: Pre-trained language models as task-specific priors," 2022, *arXiv:2210.12530*.
- [25] S. Long, A. Piché, V. Zantedeschi, T. Schuster, and A. Drouin, "Causal discovery with language models as imperfect experts," 2023, *arXiv:2307.02390*.
- [26] A. Bhattacharjee, R. Moraffah, J. Garland, and H. Liu, "Towards LLM-guided causal explainability for black-box text classifiers," in *Proc. AAAI Workshop Responsible Lang. Models*, Vancouver, BC, Canada, 2024, pp. 1–6.
- [27] Y. Zhang, Y. Zhang, Y. Gan, L. Yao, and C. Wang, "Causal graph discovery with retrieval-augmented generation based large language models," 2024, *arXiv:2402.15301*.
- [28] Z. Ji, T. Yu, Y. Xu, N. Lee, E. Ishii, and P. Fung, "Towards mitigating LLM hallucination via self reflection," in *Proc. Findings Assoc. Comput. Linguist.*, 2023, pp. 1827–1843.
- [29] A. Martino, M. Iannelli, and C. Truong, "Knowledge injection to counter large language model (LLM) hallucination," in *Proc. Eur. Semantic Web Conf.*, 2023, pp. 182–185.
- [30] T. Ban, L. Chen, X. Wang, and H. Chen, "From query tools to causal architects: Harnessing large language models for advanced causal discovery from data," 2023, *arXiv:2306.16902*.
- [31] A. Vashishtha, A. G. Reddy, A. Kumar, S. Bachu, V. N. Balasubramanian, and A. Sharma, "Causal inference using LLM-guided discovery," 2023, *arXiv:2310.15117*.
- [32] T. Feng, L. Qu, N. Tandon, Z. Li, X. Kang, and G. Haffari, "On the reliability of large language models for causal discovery," in *Proc. 63rd Annu. Meeting Assoc. Comput. Linguist. (Vol. 1: Long Papers)*, 2025, pp. 9565–9590.
- [33] M. Zečević, M. Willig, D. S. Dhami, and K. Kersting, "Causal parrots: Large language models may talk causality but are not causal," *Trans. Mach. Learn. Res.*, vol. 2023, no. 8, 2023.
- [34] Z. Wang, X. Gao, Y. Yang, X. Tan, and D. Chen, "Learning Bayesian networks based on order graph with ancestral constraints," *Knowl.-Based Syst.*, vol. 211, 2021, Art. no. 106515.
- [35] N. Harris and M. Drton, "PC algorithm for nonparanormal graphical models," *J. Mach. Learn. Res.*, vol. 14, no. 11, pp. 3365–3383, 2013.
- [36] K.-H. Cohrs, G. Varando, E. Diaz, V. Sitokostantinou, and G. Camps-Valls, "Large language models for constrained-based causal discovery," 2024, *arXiv:2406.07378*.
- [37] T. Jiralerspong, X. Chen, Y. More, V. Shah, and Y. Bengio, "Efficient causal graph discovery using large language models," 2024, *arXiv:2402.01207*.
- [38] A. Abdulaal et al., "Causal modelling agents: Causal graph discovery through synergising metadata-and data-driven reasoning," in *Proc. 12th Int. Conf. Learn. Representations*, 2023, pp. 1–52.
- [39] E. Khatibi, M. Abbasian, Z. Yang, I. Azimi, and A. M. Rahmani, "ALCM: Autonomous LLM-augmented causal discovery framework," 2024, *arXiv:2405.01744*.
- [40] M. Takayama et al., "Integrating large language models in causal discovery: A statistical causal approach," *Trans. Mach. Learn. Res.*, 2025.
- [41] N. Naik et al., "Applying large language models for causal structure learning in non small cell lung cancer," in *Proc. IEEE 12th Int. Conf. Healthcare Inform.*, 2024, pp. 688–693.
- [42] X. Liu et al., "Large language models and causal inference in collaboration: A comprehensive survey," in *Proc. Findings Assoc. Comput. Linguist.: NAACL*, 2025, pp. 7683–7699.
- [43] G. Zhou et al., "Emerging synergies in causality and deep generative models: A survey," *IEEE Trans. Artif. Intell.*, pp. 1–18, 2025.
- [44] G. Wan, Y. Wu, M. Hu, Z. Chu, and S. Li, "Large language models for causal discovery: Current landscape and future directions," in *Proc. 34th Int. Joint Conf. Artif. Intell.*, 2025, pp. 10687–10695.
- [45] S. Tong, K. Mao, Z. Huang, Y. Zhao, and K. Peng, "Automating psychological hypothesis generation with AI: When large language models meet causal graph," *Humanities Social Sci. Commun.*, vol. 11, no. 1, pp. 1–14, 2024.
- [46] E. Kıcıman, R. Ness, A. Sharma, and C. Tan, "Causal reasoning and large language models: Opening a new frontier for causality," *Trans. Mach. Learn. Res.*, 2024.
- [47] C. Zhang et al., "Understanding causality with large language models: Feasibility and opportunities," 2023, *arXiv:2304.05524*.
- [48] A. Romanou, S. Montariol, D. Paul, L. Laugier, K. Aberer, and A. Bosselut, "CRAB: Assessing the strength of causal relationships between real-world events," in *Proc. 2023 Conf. Empir. Methods Natural Lang. Process.*, 2023, pp. 15198–15216.
- [49] S. Lu, I. Bigoulaeva, R. Sachdeva, H. T. Madabushi, and I. Gurevych, "Are emergent abilities in large language models just in-context learning?," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguist.*, 2024, pp. 5098–5139.
- [50] Y. Kim, L. Guo, B. Yu, and Y. Li, "Can ChatGPT understand causal language in science claims?," in *Proc. 13th Workshop Comput. Approaches Subjectivity, Sentiment, Social Media Anal.*, 2023, pp. 379–389.
- [51] R. Tu, C. Ma, and C. Zhang, "Causal-discovery performance of chatGPT in the context of neuropathic pain diagnosis," 2023, *arXiv:2301.13819*.
- [52] M. Scutari, C. E. Graafland, and J. M. Gutiérrez, "Who learns better Bayesian network structures: Accuracy and speed of structure learning algorithms," *Int. J. Approx. Reasoning*, vol. 115, pp. 235–253, 2019.
- [53] N. K. Kitson, A. C. Constantinou, Z. Guo, Y. Liu, and K. Chobtham, "A survey of Bayesian network structure learning," *Artif. Intell. Rev.*, vol. 56, no. 8, pp. 8721–8814, 2023.
- [54] A. Li and P. Beek, "Bayesian network structure learning with side constraints," in *Proc. Int. Conf. Probabilistic Graphical Models*, 2018, pp. 225–236.
- [55] H. Dai, G. Li, and Y. Tu, "An empirical study of encoding schemes and search strategies in discovering causal networks," in *Proc. Mach. Learn.*, 2002, pp. 48–59.
- [56] I. Tsamardinos, L. E. Brown, and C. F. Aliferis, "The max-min hill-climbing Bayesian network structure learning algorithm," *Mach. Learn.*, vol. 65, pp. 31–78, 2006.
- [57] C. Yuan, B. Malone, and X. Wu, "Learning optimal Bayesian networks using A* search," in *Proc. 22nd Int. Joint Conf. Artif. Intell.*, 2011, pp. 2186–2191.
- [58] K. Zanna and A. Sano, "Uncovering bias paths with LLM-guided causal discovery: An active learning and dynamic scoring approach," in *Proc. AAAI Conf. Artif. Intell.*, vol. 40, no. 46, pp. 39567–39575, 2026.



Lyuzhou Chen received the BSc degree in mathematics and applied mathematics in 2020 from the University of Science and Technology of China, Hefei, China, where he is currently working toward the PhD degree in computer science and technology with the School of Artificial Intelligence and Data Science. His research interests include knowledge engineering and causal learning.



Xiangyu Wang (Member, IEEE) received the BSc degree from Donghua University, Shanghai, China, and the PhD degree in data science from the University of Science and Technology of China (USTC). He is currently an associate researcher with the School of Computer Science and Technology, USTC. His research interests include causal learning and machine learning.



Taiyu Ban received the BSc degree in computer science and technology in 2020 from the University of Science and Technology of China, Hefei, China, where he is currently working toward the PhD degree in computer science and technology with the School of Computer Science and Technology. His research interests include data-driven machine learning and knowledge engineering.



Derui Lyu received the BSc degree in intelligence science and technology from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2021. He is currently working toward the PhD degree in computer science and technology with the School of Computer Science and Technology, University of Science and Technology of China Hefei, China. His research interests include machine learning, causal learning, and zero-shot learning.



Xin Wang received the bachelor's degree in computer science and technology and the PhD degree from the University of Science and Technology of China (USTC) in 2019 and 2025, respectively. He is currently a postdoctoral researcher with the School of Computer Science and Technology, USTC. His research interests include causal inference, particularly counterfactual estimation, treatment effect estimation, and their applications in temporal settings and healthcare.



Qinrui Zhu received the BSc degree in computer science and technology in 2020 from the University of Science and Technology of China, Hefei, China, where he is currently working toward the PhD degree in computer science and technology with the School of Computer Science and Technology, and Laboratory for Big Data and Decision. His research interests include Big Data analysis, causal learning, and large language models.



Huanhuan Chen (Fellow, IEEE) received the BSc degree from the University of Science and Technology of China (USTC), Hefei, China, and the PhD degree in computer science from the University of Birmingham, Birmingham, U.K. He is currently a full professor with the School of Computer Science and Technology, USTC. His research interests include neural networks, Bayesian inference, and evolutionary computation. He was the recipient of the 2015 International Neural Network Society Young Investigator Award, 2012 IEEE Computational Intelligence Society Outstanding Ph.D. Dissertation Award, IEEE Transactions on Neural Networks Outstanding Paper Award (bestowed in 2011 and only one paper in 2009), and 2009 British Computer Society Distinguished Dissertations Award. He is an associate editor for *IEEE Transactions on Neural Networks and Learning Systems*, and *IEEE Transactions on Emerging Topics in Computational Intelligence*.